יום ניצוץ!
הפרויקטים הכי מבריקים בפקולטה

Modern Electromagnetic Theory and Applications

הפקולטה להנדסת חשמל ומחשבים
ע"ש אנדרו וארנה ויטרבי

Electromagnetic Metagrating Absorbers

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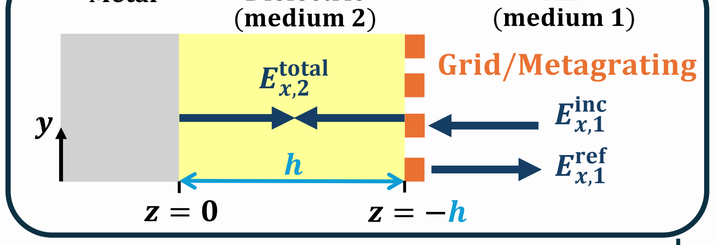
1. Introduction

- Absorption of electromagnetic waves in thin planar structures which are passive, linear, and time-invariant is desirable in many applications, ranging from radar-cross-section reduction and stealth, through solar cells and detectors for sensing, to harvesting of radiated RF power. In many of those, one would like to maximize the absorption band – a wideband device.
- In metal-backed dielectric planar structures, excited by normally incident plane-waves, an inherent trade-off between the thickness and bandwidth is found [1]. In other words, a thin absorber would limit the absorption bandwidth, and vice-versa.
- In this project, we apply rigorous semianalytical models associated with metagratings (MGs, periodic arrangements, patterned on substrates, of sparsely packed distributed loaded lines acting as secondary sources via Floquet-Bloch theorem, dictating radiation of discrete FB modes [2]), coupled with recently devised *abstract* guidelines [3], to derive closed-form and semianalytical expressions for the required constitutive distributed load parameters and metagrating (MG) geometry.

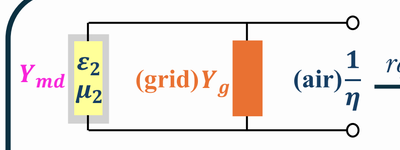
2. Theory – Analytical Formalism

A. Scattering problem

Metagratings (MGs)



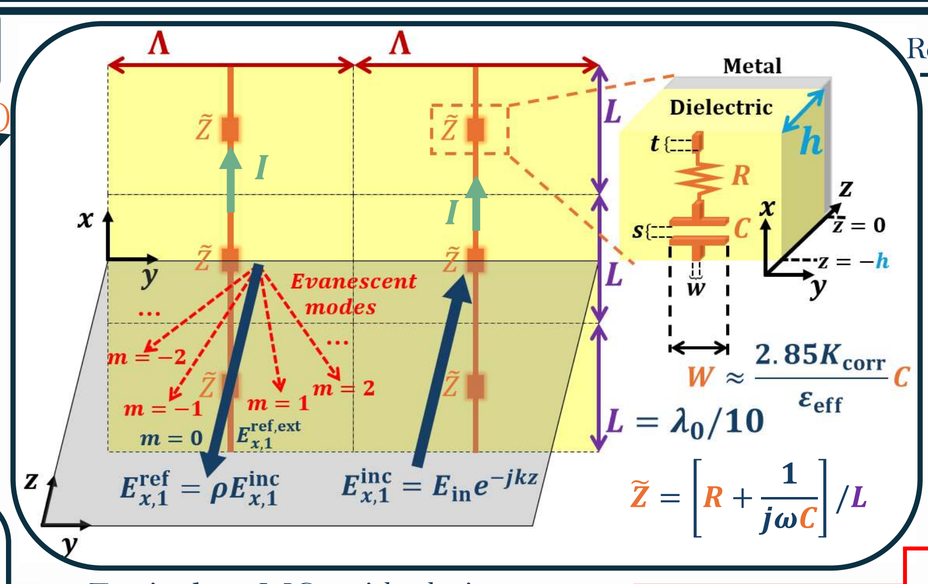
Grid: *abstract* Transmision Line Model [3]



$$\text{reflection} \longrightarrow \rho(\omega) = \frac{\frac{1}{\eta} - (Y_g + Y_{md})}{\frac{1}{\eta} + (Y_g + Y_{md})}$$

Metal-Dielectric (md) input admittance at $z = -h$:

$$Y_{md} = \frac{1}{j\eta_2 \tan(k_2 h)}$$



Resolving Fields

Propagating fields ($\Lambda < \lambda_{min}$):

$$E_{x,1}^{inc} = E_{in}e^{-jkz}$$

$$E_{x,1}^{ref} = R_0 E_{in} e^{jk(z+2h)} - \frac{\eta}{2\Lambda} I T_0 e^{jk(z+h)}$$

Floquet-Bloch reflection & wavenumber:

$$R_m = \frac{j\beta_{m,1}\tan(\beta_{m,2}h) - \beta_{m,2}}{j\beta_{m,1}\tan(\beta_{m,2}h) + \beta_{m,2}} = T_m - 1$$

$$\beta_{m,i} = \sqrt{k_i^2 - (2\pi m/\Lambda)^2}\ ;\ \alpha_{m,i} = j\beta_{m,i}$$

Ohm's Law: $E_{x,1}^{tot}(z=-h, y=0) = I\tilde{Z}$

Metagrating (MG) *reflection* coefficient:

$$\rho(\omega) = \frac{E_{x,1}^{ref}}{E_{x,1}^{inc}}\Big|_{z=-h} = \frac{R_0(\tilde{Z} - \tilde{Z}_0)}{\frac{\eta T_0^2}{2\Lambda R_0} + \tilde{Z} - \tilde{Z}_0}$$

Equivalent MG grid admittance:

$$Y_g = \left[\frac{(1-R_0)(\tilde{Z} - \tilde{Z}_0) + \frac{\eta T_0^2}{2\Lambda R_0}}{(1+R_0)(\tilde{Z} - \tilde{Z}_0) + \frac{\eta T_0^2}{2\Lambda R_0}}\right]\frac{1}{\eta} - Y_{md}$$

Perfect absorption MG load:

$$\tilde{Z}_0 = \frac{\eta T_0}{2\Lambda R_0} - j\frac{k\eta}{2\pi}\ln\left(\frac{\Lambda}{2\pi r_{eff}}\right) - j\frac{k\eta}{2\Lambda}\sum_{m\neq 0}\left\{\frac{T_m}{\alpha_{m,1}} - \frac{\Lambda}{2\pi|m|}\right\}$$

Degrees-of-freedom: $\{R, C, \Lambda, h\}$

Metagrating (MG) *absorption* coefficient: $A(\omega) = \sqrt{1 - |\rho(\omega)|^2}$

B. Rozanov's bound [1]

$$\text{Absorption Bandwidth} = \Delta\lambda < \frac{2\pi^2\mu_s^2}{|\ln\rho_0|}h$$

ρ_0 - maximally allowed reflection magnitude

We can't simultaneously decrease h and increase $\Delta\lambda$ as desired!

Compact wideband absorption is limited.

C. Grid admittance conditions [3]

Perfect absorption grid admittance ($\rho(\omega) = 0$): $Y_g^{opt} = \eta^{-1} - Y_{md}$

1. $|\rho(\omega_0)| = \rho_0$
2. $\text{Im}\{Y_g\}\big|_{\omega_0} = \text{Im}\{Y_g^{opt}\}\big|_{\omega_0}$
3. $\frac{\partial \text{Im}\{Y_g\}}{\partial\omega}\big|_{\omega_0} = \frac{\partial \text{Im}\{Y_g^{opt}\}}{\partial\omega}\big|_{\omega_0}$
4. $\frac{\partial \text{Re}\{Y_g\}}{\partial\omega}\big|_{\omega_0} = 0$

Rigorous MG model

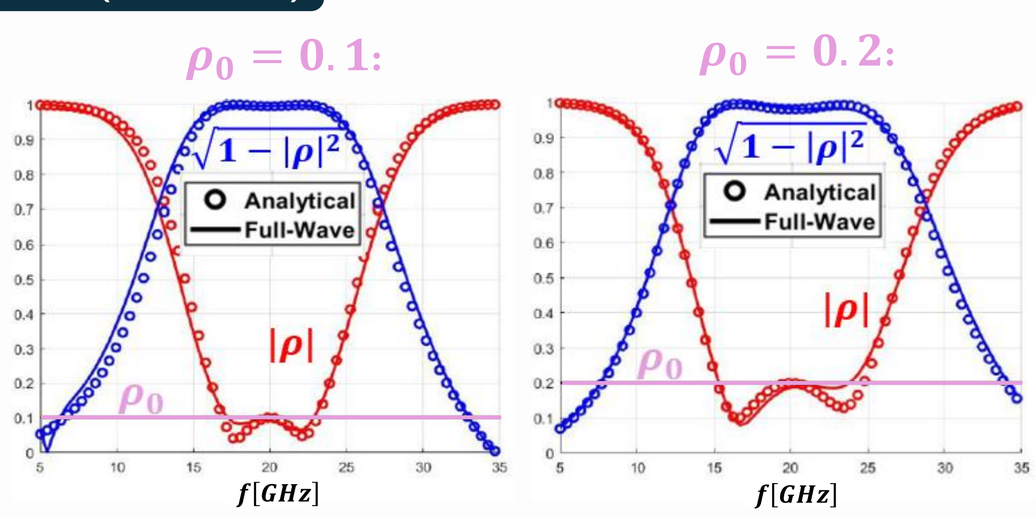
Closed-form (R, C, h) & semianalytical (Λ) formulas:

$$C = \frac{2\varepsilon_0}{k_0^2 L}\left[\frac{1}{\pi}\ln\left(\frac{\Lambda}{2\pi r_{eff}}\right) + \frac{1}{\Lambda}\sum_{m\neq 0}\left\{\frac{T_m}{\alpha_{m,1}} - \frac{\Lambda}{2\pi|m|}\right\}\right]^{-1}$$

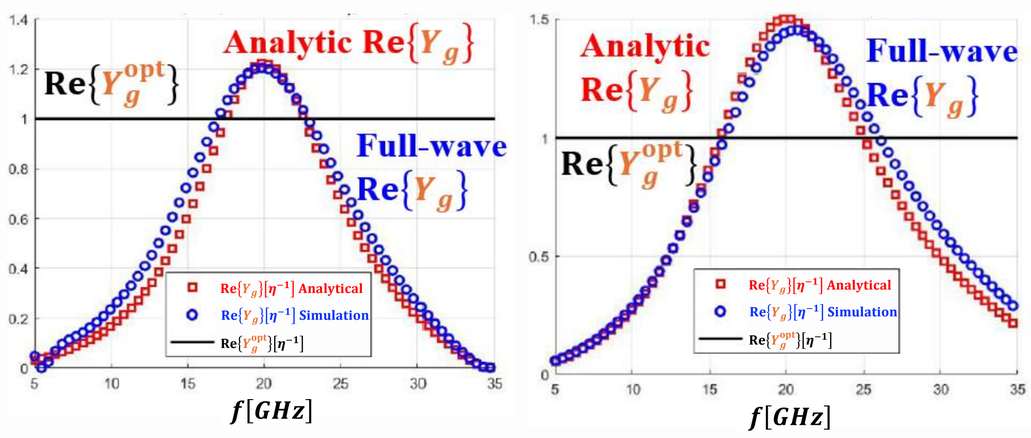
$$R = \frac{\eta L}{\Lambda}\left(\frac{1-\rho_0}{1+\rho_0}\right) \qquad h = \frac{\lambda_0}{4\sqrt{\varepsilon_{r,2}}} \qquad \Lambda = \Lambda(\omega_0, h, \rho_0)$$

3. Validation (*20GHz*)

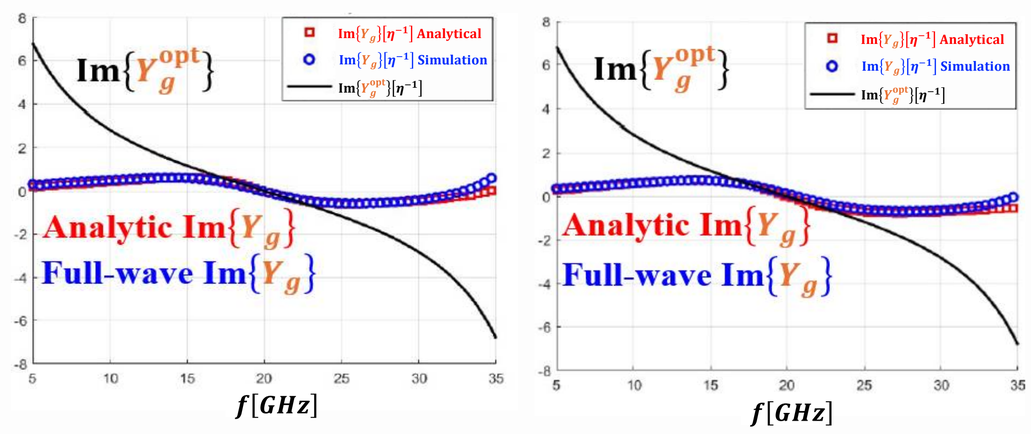
Reflection ($|\rho|$) & Absorption:



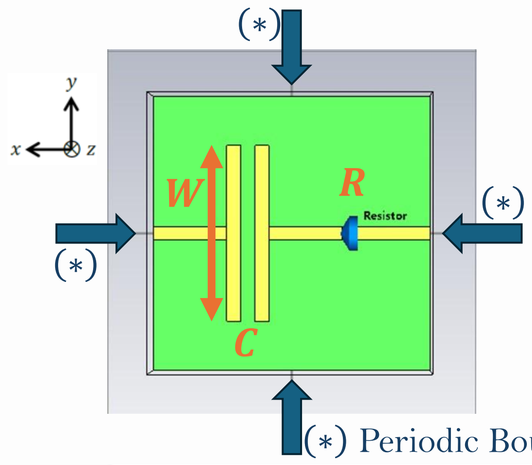
Conductance ($\text{Re}\{Y_g\}$):

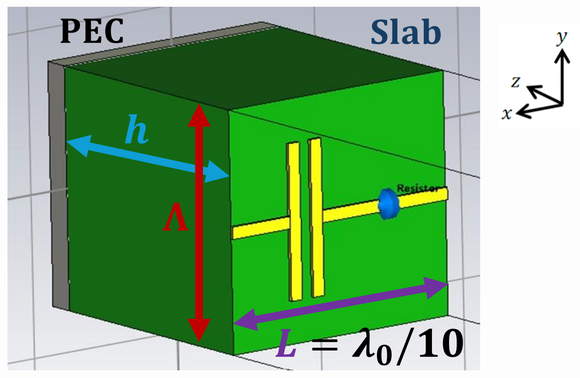


Susceptance ($\text{Im}\{Y_g\}$):



Full-Wave Validation in CST

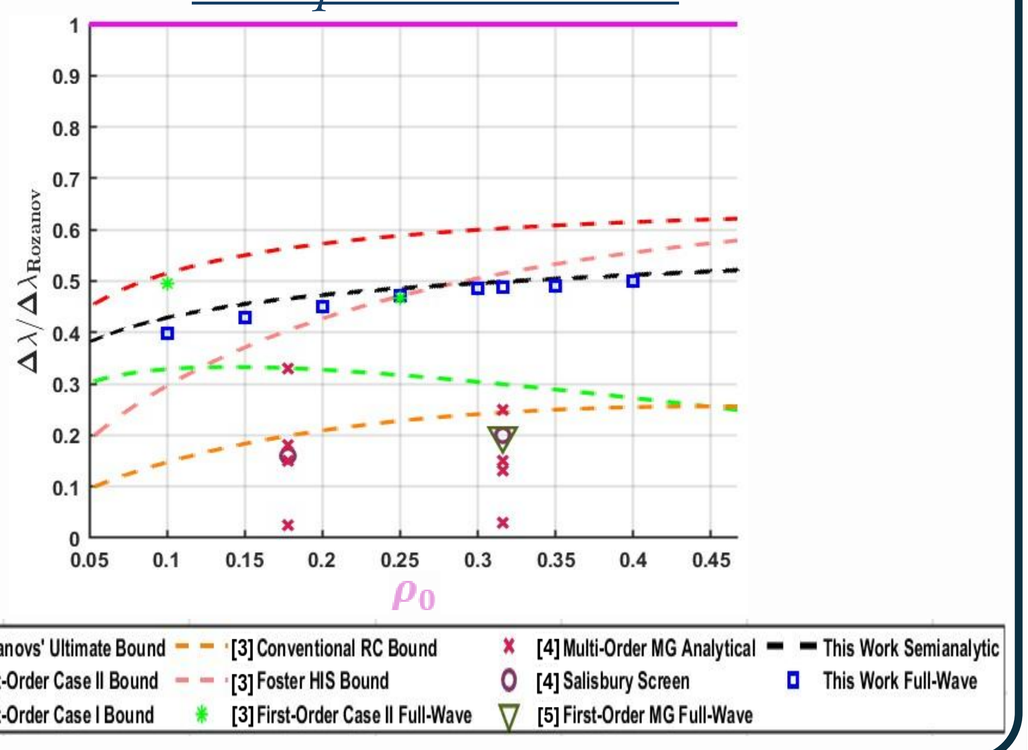




Synthesis for Rogers RO3006 ($\varepsilon_{r,2} = 6.15$) & $\rho_0 = 0.1$:

$W = 1.04[mm]$ $R = 99.06[\Omega]$ $\Lambda = 0.311\lambda_0$ $h \approx \lambda_0/10$

Absorption Bandwidth



4. References

[1] K. N. Rozanov, *IEEE Trans. Antennas Propag.*, **48**, 8 (2000).

[2] A. Epstein and O. Rabinovich, *Phys. Rev. Appl.*, **8**, 054037 (2017).

[3] P. S. Nayani, M. Moradi, P. Salami, and Y. Radi, *Nat. Commun.*, ***16***, 905 (2025).

[4] F. Boust, T. Lepetit, and S. N. Burokur, *Opt. Lett.*, ***47***, 20 (2022).

[5] Z. Tan, J. Yi, Q. Cheng, and S. N. Burokur, *IEEE Trans. Antennas Propag.*, **71**, 2 (2023).